HEAT TRANSFER LAB

Course Code: 23A03601P

B.Tech (ME) - III Year II Semester - JNTUA R23 Regulations

L-T-P-C: 0-0-3-1.5

LAB MANUAL

Name	
Roll No.	
Branch / Section	Mechanical Engineering
Academic Year	

Course Objectives

1. Understand different modes of heat transfer.
2. Gain knowledge about natural and forced convection phenomenon.
3. Estimate experimental uncertainty in measurements.

Course Outcomes

1. Explain different modes of heat transfer.
2. Identify parameters for measurement for calculating heat transfer.
3. Determine effectiveness of heat exchanger.
4. Design new equipment related to heat transfer.
5. Apply principles of heat transfer in wide application in industries.

List of Experiments

1. Overall heat transfer coefficient across the width of a composite wall
2. Thermal conductivity of a metal rod
3. Thermal conductivity of insulating powder through concentric sphere apparatus
4. Thermal conductivity of insulating material through lagged pipe apparatus
5. Efficiency of a pin fin in natural and forced convection
6. Heat transfer coefficient for a vertical cylinder in natural convection
7. Heat transfer coefficient in forced convection of air in a horizontal tube
8. Heat transfer coefficients on film and drop wise condensation apparatus
9. Effectiveness of parallel and counter flow heat exchanger
10. Study of pool boiling phenomenon and different regimes of pool boiling
11. Experiment on pool boiling
12. Emissivity of the test plate surface
13. Experiment on Stefan-Boltzmann apparatus
14. Heat transfer coefficient in fluidized bed apparatus

General Instructions and Basic Laws

- **Conduction (Fourier):** $Q = -kA\,dT/dx$. **Convection (Newton):** $Q = hA(T_s - T_\infty)$. **Radiation (Stefan-Boltzmann):** $Q = \varepsilon\sigma A(T_s^4 - T_{surr}^4)$, $\sigma = 5.67 \times 10^{-8}\ W/m^2K^4$.
- Heater input is $Q = V \times I$ (watts). Use temperatures in kelvin for radiation formulae. Wait for **steady state** (thermocouple readings unchanged for 5-10 min) before noting readings.
- Switch on the dimmerstat slowly from zero, never exceed the specified voltage, and switch off the heater before draining water or opening the setup.

Estimation of Experimental Uncertainty

For a result $R = f(x_1, x_2, ...)$ with measurement uncertainties δx_i, the uncertainty is $\delta R = \sqrt{[\,(\partial R/\partial x_1\,\delta x_1)^2 + (\partial R/\partial x_2\,\delta x_2)^2 + ...\,]}$. For a product $R = x^a y^b$: $\delta R/R = \sqrt{[(a\,\delta x/x)^2 + (b\,\delta y/y)^2]}$. Example: for $h = Q/(A\,\Delta T)$ with $Q = VI$, $\delta h/h = \sqrt{[(\delta V/V)^2 + (\delta I/I)^2 + (\delta A/A)^2 + (\delta \Delta T/\Delta T)^2]}$. Report each final result as value ± uncertainty.

Experiment 1: Overall Heat Transfer Coefficient across a Composite Wall

Aim

To determine the overall heat transfer coefficient and the interface temperatures of a composite wall made of slabs of different materials.

Apparatus

Composite wall setup (MS, aluminium, bakelite/wood slabs clamped in series) with a central heater, dimmerstat, voltmeter, ammeter, thermocouples with selector switch and digital temperature indicator.

Theory and Formulae

- For slabs in series, thermal resistance of each slab $R = L/(kA)$; total $R_{total} = L_1/(k_1A) + L_2/(k_2A) + L_3/(k_3A)$.
- Heat flow: $Q = (T_1 - T_n)/R_{total} = VI$ (neglecting side losses).
- Overall heat transfer coefficient: $U = Q / [A (T_{hot} - T_{cold})]$, and $U = 1/(1/h_i + \Sigma L/k + 1/h_o)$ when surface films are included.

Procedure

1. Note slab thicknesses and area A.
2. Switch on the supply and set the dimmerstat to a suitable voltage.
3. Wait for steady state; record V, I and all thermocouple temperatures.
4. Repeat for another heater input.
5. Calculate Q, U and compare the interface temperatures with the theoretical profile.

Observation Table

Sl.	V (V)	I (A)	T1..T6 (C)	Q = VI (W)	U (W/m2K)

Result

The overall heat transfer coefficient of the composite wall was found to be _____ W/m^2K.

Viva Questions

1. Define thermal resistance.
2. Why are slabs in series assumed to carry the same heat flow?
3. What is the contact resistance?

Experiment 2: Thermal Conductivity of a Metal Rod

Aim

To determine the thermal conductivity of a metal rod by the axial conduction method.

Apparatus

Metal rod (brass/copper/aluminium) with a heater at one end, water-cooled or free end, asbestos/glass-wool lagging, thermocouples along the length, voltmeter, ammeter, dimmerstat.

Theory and Formulae

- One-dimensional steady conduction (Fourier law): $Q = -kA\,dT/dx$, so $k = Q\,x / [A\,(T_1 - T_2)]$ between two points a distance x apart.
- Heat input $Q = VI$ (W) corrected for lagging losses; $A = \pi d^2/4$.
- k obtained from temperature gradient: $k = Q / (A \cdot dT/dx)$.

Procedure

1. Measure rod diameter d and the thermocouple spacing.
2. Switch on the heater and the cooling water (if provided).
3. Allow steady state; note V, I and temperatures T1 to T6 along the rod.
4. Plot T against x; obtain slope dT/dx and compute k.
5. Repeat for another voltage.

Observation Table

Sl.	V (V)	I (A)	T1..T6 (C)	dT/dx (C/m)	k (W/mK)

Result

The thermal conductivity of the rod = _____ W/mK (compare with the handbook value).

Viva Questions

1. State Fourier law.
2. Why is the rod lagged?
3. Typical k of copper and aluminium?

Experiment 3: Thermal Conductivity of Insulating Powder - Concentric Sphere Apparatus

Aim

To determine the thermal conductivity of an insulating powder filled between two concentric spheres.

Apparatus

Two concentric copper/aluminium spheres (inner sphere with heater, outer sphere), insulating powder filled in the annulus, thermocouples on the inner and outer surfaces, dimmerstat, voltmeter, ammeter.

Theory and Formulae

- For steady radial conduction through a spherical shell: $Q = 4\pi k\, r_1\, r_2\, (T_1 - T_2) / (r_2 - r_1)$.
- Hence $k = Q\,(r_2 - r_1) / [4\pi\, r_1\, r_2\, (T_1 - T_2)]$, where $Q = VI$, T_1 = mean inner sphere temperature, T_2 = mean outer sphere temperature.

Procedure

1. Note r1 and r2.
2. Switch on the heater and set the input using the dimmerstat.
3. Wait for steady state (it may take 1.5 to 2 hours) and record V, I and temperatures.
4. Calculate the mean T1 and T2 and then k.

Observation Table

Sl.	V (V)	I (A)	T1 mean (C)	T2 mean (C)	k (W/mK)

Result

The thermal conductivity of the insulating powder = _____ W/mK.

Viva Questions

1. Why is a sphere used for powders?
2. Derive the heat flow equation for a hollow sphere.

Experiment 4: Thermal Conductivity of Insulating Material - Lagged Pipe Apparatus

Aim

To determine the thermal conductivity of the lagging (insulating) material wrapped over a pipe.

Apparatus

Central heated pipe with concentric layers of lagging material, thermocouples on the pipe surface and between layers, voltmeter, ammeter, dimmerstat.

Theory and Formulae

- For radial conduction through a hollow cylinder of length L: $Q = 2\pi kL\,(T_1 - T_2) / \ln(r_2/r_1)$.
- Thus $k = Q \ln(r_2/r_1) / [2\pi L\,(T_1 - T_2)]$, with $Q = VI$.
- For two layers in series: $Q = 2\pi L\,(T_1 - T_3) / [\ln(r_2/r_1)/k_1 + \ln(r_3/r_2)/k_2]$.

Procedure

1. Note the radii of the pipe and each layer and the heated length L.
2. Switch on the heater and adjust the input.
3. At steady state, note V, I and temperatures.
4. Calculate k of each layer.

Observation Table

Sl.	V (V)	I (A)	T1 (C)	T2 (C)	k (W/mK)

Result

The thermal conductivity of the lagging material = _____ W/mK.

Viva Questions

1. What is the critical radius of insulation?
2. Why does a lagged pipe lose less heat?

Experiment 5: Efficiency of a Pin Fin in Natural and Forced Convection

Aim

To determine the temperature distribution, heat transfer coefficient and efficiency of a pin fin under natural and forced convection.

Apparatus

Pin fin (brass) in a rectangular duct with a heater at the base, blower with air flow control, orifice meter and manometer, thermocouples along the fin, dimmerstat, voltmeter, ammeter.

Theory and Formulae

- Fin parameter $m = \sqrt{(hP / kA_c)}$, with $P = \pi d$, $A_c = \pi d^2/4$.
- For a fin with insulated tip: temperature profile $(T - T_\infty)/(T_b - T_\infty) = \cosh m(L - x)/\cosh(mL)$; heat flow $Q = \sqrt{(hPkA_c)}\,(T_b - T_\infty)\tanh(mL)$.
- **Fin efficiency** $\eta = \tanh(mL)/(mL)$ = actual heat transfer / heat transfer if the whole fin were at base temperature = $Q_{fin} / [hPL\,(T_b - T_\infty)]$.
- Fin effectiveness = $Q_{fin} / [hA_c(T_b - T_\infty)]$.

Procedure

1. Measure fin dimensions; switch on the heater (blower off) for the natural convection case.
2. At steady state, record temperatures T1 to T5 along the fin and the ambient temperature.
3. Switch on the blower, set the air velocity using the manometer reading and repeat for forced convection.
4. Compute h from the observed profile (or from correlations), then m, mL and efficiency.

Observation Table

Condition	T base (C)	T tip (C)	T amb (C)	h (W/m2K)	Efficiency

Result

The efficiency of the pin fin was found to be _____ (natural) and _____ (forced). Efficiency is higher for natural convection because h is lower.

Viva Questions

1. What is a fin and why is it used?
2. Define fin efficiency and effectiveness.
3. Effect of h on efficiency?

Experiment 6: Heat Transfer Coefficient for a Vertical Cylinder in Natural Convection

Aim

To determine the average natural convection heat transfer coefficient for a vertical heated cylinder and compare it with the theoretical value.

Apparatus

Vertical cylinder (brass) with an internal heater placed in an enclosure open to the atmosphere, thermocouples on the surface and ambient, dimmerstat, voltmeter, ammeter.

Theory and Formulae

- Experimental $h = Q / [A(T_s - T_\infty)]$, where $Q = VI$ (radiation loss may be subtracted), $A = \pi dL$.
- Film temperature $T_f = (T_s + T_\infty)/2$; $\beta = 1/T_f$ (K); fluid properties (ν, k, Pr) at T_f.
- $Gr = g\beta(T_s - T_\infty)L^3/\nu^2$; $Ra = Gr\ Pr$.
- Theoretical: $Nu = 0.59\ Ra^{0.25}$ ($10^4 < Ra < 10^9$), and $h = Nu\ k/L$.

Procedure

1. Note dimensions; switch on the heater and set the voltage.
2. Wait for steady state; record V, I, surface temperatures T1 to T5 and the ambient temperature.
3. Repeat for two other power inputs.
4. Calculate h (experimental) and h (theoretical) and compare.

Observation Table

Sl.	V (V)	I (A)	Ts mean (C)	T amb (C)	h exp (W/m2K)	h theory

Result

The average natural convection heat transfer coefficient = _____ W/m^2K (experimental) and _____ (theoretical).

Viva Questions

1. What are the Grashof and Prandtl numbers?
2. Why does the boundary layer thicken with height?

Experiment 7: Heat Transfer Coefficient in Forced Convection of Air in a Horizontal Tube

Aim

To determine the forced convection heat transfer coefficient for air flowing through a heated horizontal tube.

Apparatus

Horizontal tube with a heater winding (nichrome) and lagging, blower, orifice meter with U-tube manometer, thermocouples on the tube wall and at the air inlet and outlet, dimmerstat, voltmeter, ammeter.

Theory and Formulae

- Air velocity from the orifice meter: $V_a = C_d\, a_o \sqrt{(2\, g\, H\, \rho_w/\rho_a)} / A_{pipe}$, with H = manometer difference of water column.
- Heat gained by air: $Q = m\blacksquare\, c_p\, (T_{out} - T_{in}) \approx VI$. $h = Q / [\pi dL\, (T_w - T_{mean,air})]$.
- Nu = hd/k; Re = Vd/ν. Dittus-Boelter (heating): $Nu = 0.023\, Re^{0.8}\, Pr^{0.4}$ for Re > 10,000.

Procedure

1. Note tube diameter, length and orifice diameter.
2. Start the blower and set a flow rate; switch on the heater.
3. At steady state record V, I, manometer reading, wall and air temperatures.
4. Repeat for two other flow rates.
5. Compare experimental Nu with Dittus-Boelter.

Observation Table

Sl.	H (m)	V (m/s)	Tw (C)	T mean air (C)	h (W/m2K)	Nu

Result

The forced convection heat transfer coefficient = _____ W/m^2K. h increases with velocity.

Viva Questions

1. Define Reynolds and Nusselt numbers.
2. Why is the Dittus-Boelter correlation used?

Experiment 8: Film and Drop Wise Condensation

Aim

To determine the heat transfer coefficients for film-wise and drop-wise condensation of steam on vertical tubes.

Apparatus

Condensation apparatus with a steam generator (boiler), two vertical tubes (one chrome/gold plated or with promoter for drop-wise, other plain copper for film-wise), cooling water supply with rotameters, thermocouples, pressure gauge, measuring jar.

Theory and Formulae

- Heat given by steam = heat gained by cooling water: $Q = m_w c_p (T_{wo} - T_{wi})$.
- $h = Q / [A (T_{sat} - T_w)]$, $A = \pi d L$.
- Film condensation (Nusselt, vertical surface): $h = 0.943 [\rho_l(\rho_l - \rho_v) g h_{fg} k_l^3 / (\mu_l L \Delta T)]^{1/4}$.
- Drop-wise condensation gives a much higher h (about 5 to 20 times) because the surface is not covered by an insulating liquid film.

Procedure

1. Fill the boiler with water and heat to generate steam; wait for steady pressure.
2. Start the cooling water flow through each tube at a fixed rate.
3. At steady state, note steam pressure/temperature, wall temperatures, water inlet and outlet temperatures and flow rate.
4. Calculate h for the film and drop-wise tubes and compare.

Observation Table

Type	T sat (C)	T wall (C)	Tw in / out (C)	Q (W)	h (W/m2K)

Result

h(film) = _____ W/m^2K and h(drop-wise) = _____ W/m^2K; drop-wise condensation gives the higher coefficient.

Viva Questions

1. Why is drop-wise condensation better?
2. Why are non-condensable gases harmful?

Experiment 9: Effectiveness of Parallel and Counter Flow Heat Exchanger

Aim

To determine the LMTD, overall heat transfer coefficient and effectiveness of a double-pipe heat exchanger in parallel and counter flow modes.

Apparatus

Concentric tube heat exchanger, hot water geyser/tank, cold water supply, rotameters or measuring jar, valves for changing flow direction, thermocouples at inlets and outlets of both fluids.

Theory and Formulae

- Heat given by hot fluid: $Q_h = m_h c_{ph} (T_{hi} - T_{ho})$; heat gained by cold fluid: $Q_c = m_c c_{pc} (T_{co} - T_{ci})$; Q = average of the two.
- LMTD = $(\Delta T_1 - \Delta T_2) / \ln(\Delta T_1/\Delta T_2)$. Parallel: $\Delta T_1 = T_{hi} - T_{ci}$, $\Delta T_2 = T_{ho} - T_{co}$. Counter: $\Delta T_1 = T_{hi} - T_{co}$, $\Delta T_2 = T_{ho} - T_{ci}$.
- U = Q / (A · LMTD).
- **Effectiveness** $\varepsilon = Q / Q_{max} = Q / [C_{min}(T_{hi} - T_{ci})]$, where $C = mc_p$.
- NTU = UA/C_{min}, $C_r = C_{min}/C_{max}$. Parallel: $\varepsilon = [1 - e^{-NTU(1+Cr)}]/(1 + C_r)$. Counter: $\varepsilon = [1 - e^{-NTU(1-Cr)}]/[1 - C_r e^{-NTU(1-Cr)}]$.

Procedure

1. Start the hot water and cold water flow; set the valves for parallel flow.
2. At steady state, record flow rates and the four temperatures.
3. Reverse the cold water direction for counter flow and repeat.
4. Calculate LMTD, U and effectiveness for both modes and compare.

Observation Table

Mode	T hi / ho (C)	T ci / co (C)	Q (W)	LMTD (C)	Effectiveness

Result

Effectiveness: parallel flow = _____, counter flow = _____. Counter flow gives higher effectiveness and LMTD for the same inlet conditions.

Viva Questions

1. Why is counter flow more effective?
2. What is NTU?
3. Can T_{co} exceed T_{ho} in parallel flow?

Experiment 10: Study of Pool Boiling Phenomenon and Regimes

Aim

To study pool boiling and its different regimes.

Theory

Boiling is convective heat transfer with phase change. In **pool boiling** the liquid is stagnant and motion is due to natural convection and bubble movement. The boiling curve plots heat flux q" against excess temperature $\Delta T_e = T_s - T_{sat}$.

Regime	Approx. ΔT_e (water, 1 atm)	Description
Natural convection boiling	below about 5 C	Liquid is slightly superheated; no bubbles; heat transfer by free convection.
Nucleate boiling	about 5 to 30 C	Bubbles form at nucleation sites, rise and cause strong mixing; h is very high. Ends at the **critical heat flux (CHF)** point.
Transition boiling	about 30 to 120 C	Unstable partial vapour film; heat flux falls as ΔT_e rises. Minimum point is the Leidenfrost point.
Film boiling	above about 120 C	Continuous stable vapour film covers the surface; radiation contributes; heat flux rises again.

Burnout: in heat-flux controlled systems (electric heaters) exceeding the CHF makes the surface temperature jump to the film-boiling value, which can melt the heater. Zuber CHF for large flat surfaces: $q''_{max} = 0.131\, h_{fg}\, \rho_v^{1/2}\, [\sigma\, g\, (\rho_l - \rho_v)]^{1/4}$ (about 1.1 MW/m^2 for water at 1 atm).

Result

The regimes of pool boiling were studied and the boiling curve was understood.

Experiment 11: Experiment on Pool Boiling

Aim

To determine the boiling heat transfer coefficient and plot the boiling curve for water boiling on an electrically heated wire/tube.

Apparatus

Glass vessel filled with distilled water, nichrome/stainless steel wire (or tube) heater, auxiliary heater, thermocouple on the surface, thermometer for the liquid, dimmerstat, voltmeter, ammeter.

Theory and Formulae

- Heat flux $q'' = VI/A$, where $A = \pi dL$ of the wire. Excess temperature $\Delta T_e = T_w - T_{sat}$.
- Boiling heat transfer coefficient $h = q'' / \Delta T_e = VI / [A (T_w - T_{sat})]$.
- Nucleate boiling correlation (Rohsenow) may be used for comparison.

Procedure

1. Fill the vessel with distilled water and heat it to saturation (100 C) using the auxiliary heater.
2. Switch on the wire heater and increase the voltage in small steps.
3. At each step note V, I and the wire temperature after it steadies; observe bubble formation.
4. Plot q'' against ΔT_e on a log-log scale. Stop before burnout.

Observation Table

Sl.	V (V)	I (A)	T wire (C)	Heat flux (W/m2)	h (W/m2K)

Result

The boiling curve was plotted; h = _____ W/m^2K in the nucleate regime.

Viva Questions

1. What is the critical heat flux?
2. What is the Leidenfrost point?
3. What is burnout?

Experiment 12: Emissivity of the Test Plate Surface

Aim

To determine the emissivity of a test plate surface by comparing it with a black plate.

Apparatus

Two identical circular plates (one blackened, one test plate) heated by separate heaters inside a closed chamber, two dimmerstats, voltmeters, ammeters, thermocouples on both plates and the enclosure.

Theory and Formulae

- Radiation heat loss from a plate: $Q = \varepsilon\sigma A\,(T_s^{\,4} - T_{surr}^{\,4})$, temperatures in kelvin.
- With both plates at the same temperature, the heat inputs are proportional to emissivities: $\varepsilon_{test} = \varepsilon_b\,(W_{test} / W_b)$, where $W = VI$ and $\varepsilon_b \approx 1$ for the black plate.
- Alternatively $\varepsilon = Q / [\sigma A\,(T_s^{\,4} - T_{surr}^{\,4})]$ (neglecting convection in a vacuum/enclosed chamber).

Procedure

1. Switch on both heaters; adjust the dimmerstats.
2. Adjust the inputs until both plates reach the same steady temperature.
3. Note V and I for each plate.
4. Calculate the emissivity of the test plate.

Observation Table

Plate	V (V)	I (A)	W = VI (W)	Ts (C)	Emissivity

Result

The emissivity of the test plate surface = _____.

Viva Questions

1. Define emissivity and a black body.
2. Value of emissivity of polished metals?

Experiment 13: Stefan-Boltzmann Apparatus

Aim

To determine the Stefan-Boltzmann constant.

Apparatus

Water-heated (or electrically heated) hemispherical enclosure with a copper test disc at the centre, thermocouples on the disc and enclosure, stopwatch, voltmeter/ammeter for the heater.

Theory and Formulae

- Net radiation exchange between the enclosure (T_h) and the small disc (T_d): $Q = \sigma A (T_h^4 - T_d^4)$, emissivity of the blackened surfaces taken as 1.
- Heat gained by the disc: $Q = m c_p (dT_d/dt)$, measured as the disc temperature rises from its initial value.
- Hence $\sigma = m c_p (dT_d/dt) / [A (T_h^4 - T_d^4)]$. Accepted value 5.67×10^{-8} W/m^{2}K^4.

Procedure

1. Heat the hemisphere to a steady temperature T_h.
2. Keep the disc at room temperature, then insert it at the centre of the hemisphere.
3. Record the disc temperature every 10 s for about 3 min.
4. Plot T_d against time; find dT_d/dt at $t = 0$ and calculate σ.

Observation Table

Time (s)	Td (C)	Td (K)	Th (K)	dT/dt (K/s)	sigma

Result

The Stefan-Boltzmann constant was found to be _____ W/m^{2}K^4.

Viva Questions

1. State the Stefan-Boltzmann law.
2. Why must temperatures be in kelvin?

Experiment 14: Heat Transfer Coefficient in Fluidized Bed Apparatus

Aim

To determine the heat transfer coefficient between a heated surface and a fluidized bed of particles.

Apparatus

Vertical column filled with sand/glass beads, air blower with flow control, orifice meter and manometer, heater (immersed cartridge/wall heater), thermocouples on the heater surface and bed, bed pressure drop manometer, dimmerstat.

Theory and Formulae

- $h = Q / [A (T_s - T_{bed})]$ with $Q = VI$.
- At **minimum fluidization** the pressure drop across the bed balances the bed weight: $\Delta P = (1 - \varepsilon)(\rho_p - \rho_f) g H$.
- h rises sharply once fluidization starts (particle motion scrubs the boundary layer), reaches a maximum slightly above the minimum fluidization velocity and then falls or levels off as bubbles increase.

Procedure

1. Fill the column to a known static height and note the particle size.
2. Start the air flow at a low rate; note the bed pressure drop and heater readings after steady state.
3. Increase the flow in steps, through the fixed-bed, minimum fluidization and fluidized states.
4. Plot ΔP and h against air velocity.

Observation Table

Air flow (m3/s)	Velocity (m/s)	Delta P (mm)	Ts (C)	T bed (C)	h (W/m2K)

Result

The heat transfer coefficient in the fluidized bed = _____ W/m^2K and the minimum fluidization velocity = _____ m/s.

Viva Questions

1. What is fluidization?
2. Why is h higher in a fluidized bed than in a packed bed?

Virtual Lab Simulations

The syllabus lists these online simulations (links as given in the syllabus, Amrita Virtual Labs and the BNMIT heat transfer virtual lab):

1. Determination of thermal conductivity of a metal rod
2. Natural convection heat transfer
3. Heat transfer by radiation
4. Heat transfer by conduction
5. The study of phase change
6. Black body radiation: determination of Stefan's constant
7. Newton's law of cooling
8. Lee's disc apparatus
9. Thermocouple - Seebeck effect

Newton's law of cooling: $dT/dt = -k\,(T - T_\infty)$ so $T = T_\infty + (T_0 - T_\infty)\,e^{-kt}$. Lee's disc: $k = Q\,d / [A\,(T_1 - T_2)]$, used for poor conductors.